



storage, depreciation, and missed opportunities loom large, casting a shadow over profitability and growth prospects.

Electronics technology is rapidly evolving, raising the fear of obsolescence of components which could today be the *creme de la creme* of electronics innovation. With rapid technological advancements, short product life cycles, increasing competition, price pressures, and environmental regulations, electronics manufacturing is no fool's errand. So naturally, every penny saved counts!

Tracing the treasure

So how did this excess stock come into existence in an EMS? For that, one needs to understand the basic categories of on demand production that an EMS encounters—prototypes, pre-series, and series. Shamim Alam, Assistant Manager - Sourcing, Cyient DLM, told EFY, "Prototype demand is obviously lower, with pre-series falling in between, and series being the highest. This determines the inventory strategy of the EMS."

Elaborating on the distinct types of inventory management, Alam continued, "For prototypes, EMS manufacturers mostly rely on just-in-time inventory management, with some buffer for potential failures or losses. You can't afford to keep excess inventory because components might get upgraded, rendering them obsolete. For pre-series and series demand, a good inventory management ratio involves maintaining around 30% of the required components in stock."

"Your inventory management strategy should align with the type of demand you are catering to and the current market scenario," says Alam.

In an EMS company, there are two approaches to stocking inven-

Specs: Important?

This revelation makes it even more pertinent to have a domestic electronics component manufacturing ecosystem in place. In 2020, the government of India introduced the Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECES) to offset the disability for domestic manufacturing of electronic components and semiconductors, providing financial incentives up to 25% on capital expenditure for identified electronic goods involved in high value-added manufacturing. This capex includes plant, machinery, equipment, utilities, technology, and R&D, for applications up to 3 years. As of October 2023, the IT Ministry has reportedly accepted 38 applications, representing a proposed investment of ₹114.85 billion (11,485 crore), with committed incentives of ₹15.74 billion (1,574 crore) under the SPECES scheme. However, only about ₹3.65 billion (365 crore) had been disbursed to six applications till that time. In December 2023, S Krishnan, Secretary, Ministry of Electronics and IT (MeitY), said that the government was working on the second version of the incentive scheme.

tory—made-to-order and made-to-stock. Manufacturing is primarily driven by demand rather than producing goods and then trying to sell them in the market. If the guidelines are clear, one wonders what might be filling up the inventory shelves of EMS companies?

For this we need to travel back in time to the year when a pandemic brought the entire world to a stand-still, reducing the production capacity of the entire electronics components supply chain. Faced with widespread shortages and rising demand, manufacturers started hoarding components to increase their inventory build to meet customer demands, and a new system was formed.

The conundrum of components

Procurement of components is based on current demand from various customers. Goods entering the factory are always accompanied by a purchase order. The problem arises when the ordering is not aligned with actual utilisation, resulting in ad-hoc purchases, which ends up as unused or partially utilised inventory that eventually becomes slow-moving stock and may end up as dead stock. Now the question arises, why is the ordering process failing to prevent this?

The answer lies in the material

that is being ordered. Unlike regular orders, electronics components are not procured on a demand basis. Usually, most electronics component suppliers have a minimum order quantity (MOQ). Nandini reflects back on the electronics component shortage. "We were trying to maintain a stock of long lead time items as per our projects, but we could never get a forecast for more than 3-4 months whereas the lead times were like 52-74 weeks," she says.

She elaborates, "If I have a forecast and an ordering pattern which is already known to me, I will start ordering for the next six months. The problem arises when I don't have all the components at the same time to make a product. So, if one product needs 200 electronics components, even if I have 190 in stock, I will not be able to produce the product because I need all 200. So, at any given point in time, we will have a huge stock of mismatched inventory."

To get the manufacturing process back on track, most electronics manufacturers resort to buying the missing components to meet the monthly demands, which ends up in a mismatch in the future.

With most component manufacturers located outside India, domestic electronics manufacturers lack any control over the acquisition and